

Lung Cancer: An Approach to a Case of Haemoptysis

Dr. Essam Kotb, Associate Professor, Internal Medicine | Source: 8-Lung_Cancer_Updated.pdf (35 slides) · Refs: Step-Up 5th ed.; Harrison 20th ed.; AMBOSS; Cecil 10th ed.; First Aid IM Boards 4th ed.; Osmosis

Legend: ★ = high-yield · [added] = not on slides. Images marked "Slide N" come from the lecture; "Added diagram" = drawn for these notes.

🔗 **Bridge from the Pleura lecture (same lecturer):** malignancy was a cause of **exudative effusion** and **haemothorax**, and **lymphocyte-rich exudate** suggested TB or cancer. Lung cancer was also a cause of **lung abscess** (bronchial obstruction) and **clubbing in COPD** (a red flag). This lecture covers the cancer itself and the symptom it often presents with: **haemoptysis**.

Lung cancer types: where it sits, what it secretes, how it behaves

ADENOCARCINOMA NSCLC · alveolar type II cells	SQUAMOUS CELL NSCLC · basal epithelial cells	LARGE CELL NSCLC · various epithelial	SMALL CELL (SCLC) ~15% · neuroendocrine
PERIPHERAL Most common type Women, NON-smokers Arises in scars Best prognosis Paraneoplastic: Hypertrophic osteo-arthropathy (clubbing) Non-bacterial verrucous endocarditis	CENTRAL Strong smoking link CAVITATES (from a hilar bronchus) Paraneoplastic: PTHrP → HYPERCALCAEMIA (poor prognosis)	PERIPHERAL Strong smoking link Early metastases Poor chemo response Poor prognosis Paraneoplastic: GYNAECOMASTIA (hCG)	CENTRAL Strong smoking link Very aggressive, early metastases Staged LIMITED vs EXTENSIVE (not TNM) Paraneoplastic: ACTH → Cushing SIADH → low Na ⁺ Lambert-Eaton SVC syndrome (more often)

Mnemonic: **S**quamous & **S**mall = **S**entral (central) · All cancers: anaemia, DIC, thrombocytosis, dermatomyositis
Pancoast tumour (apical, usually NSCLC) → Horner (ptosis, miosis, anhidrosis), C8-T2 arm pain, hoarseness

Added diagram: the four lung cancer types

1. Title / Objectives

Domain	Objective	Likely tested as
Knowledge	★ Epidemiology and risk factors: smoking, radon , asbestos , other carcinogens	Smoking 80-90%; radon #2
Knowledge	★ Histological types: location and cell of origin	Central vs peripheral; adeno in non-smokers
Knowledge	★ Presentation: pulmonary, constitutional, local invasion, metastatic	SVC, Pancoast, Horner
Knowledge	★★ Paraneoplastic syndromes by subtype	SIADH/ACTH/Lambert-Eaton = SCLC; PTHrP = squamous
Knowledge	★ TNM (NSCLC) vs limited/extensive (SCLC)	Staging system choice
Knowledge	Haemoptysis differential and work-up; low-dose CT screening	Screening criteria

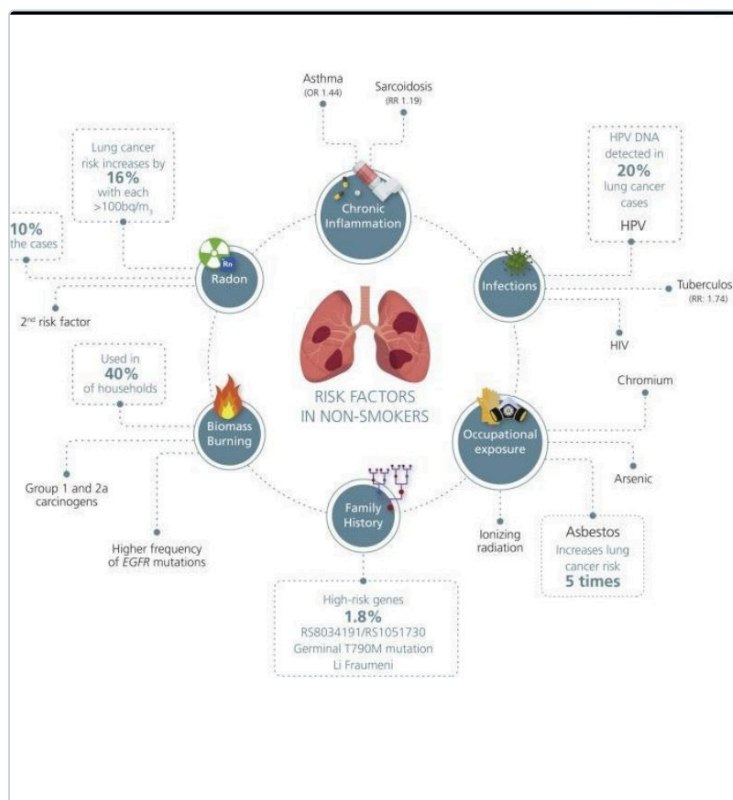
Domain	Objective	Likely tested as
Skills	Interpret CXR/CT; solitary nodule algorithm ; massive vs non-massive haemoptysis ; tissue diagnosis and staging; treatment by stage/type; resuscitation in massive haemoptysis	Case vignettes
Values	Multidisciplinary care, guidelines, screening	

Flow: Risk factors → Types → Staging (TNM, SCLC) → Signs & symptoms (local invasion: SVC, Pancoast, Horner; pulmonary; extrapulmonary; metastases; paraneoplastic) → Diagnostics → Management → Screening → Solitary nodule → Haemoptysis (history, exam, DDX, work-up, massive) → Summary → Cases.

2. Main Content (Lecture Order)

2.1 Risk Factors (Slide 5)

- ★★ **Smoking:** 80-90% of lung cancers; related to **pack-years**
- ★ **Occupational / environmental:** **passive smoking** · ★ **radon (2nd leading cause)** and uranium · ★ **asbestos** · arsenic, silica · air pollution
- Family history** (genetic predisposition)
- Others:** **pulmonary scarring, previous radiation, pulmonary fibrosis**, chronic infection (**TB, HIV**)

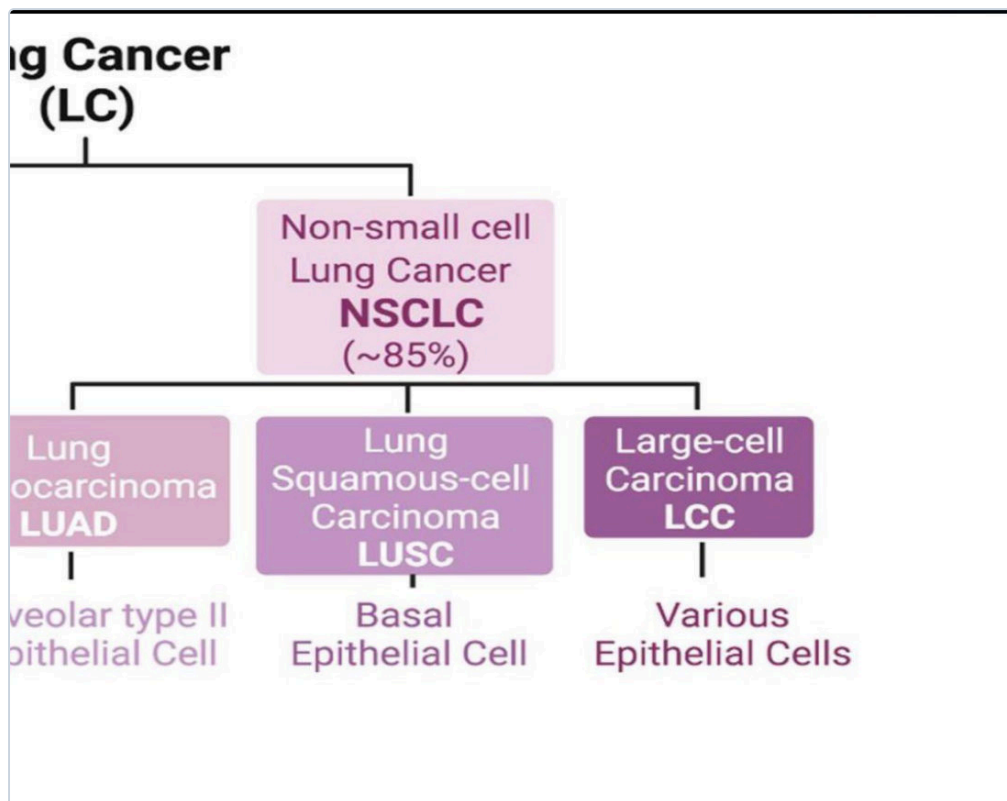


Slide 5: Lung cancer risk factors

2.2 Types (Slides 6-7)

	SCLC	NSCLC
Share	~15%	★ ~85%

	SCLC	NSCLC
Subtypes	Small cell (neuroendocrine)	Adenocarcinoma (alveolar type II cells) · Squamous (basal epithelial cells) · Large cell (various epithelial cells)



Slide 6: Lung cancer classification and cells of origin

★★★★ Type comparison (Slide 7):

Type	★ Location	★ Key features
Adenocarcinoma	★ Peripheral	★ Most common primary lung cancer ; ★ more common in women and non-smokers ; hypertrophic osteoarthropathy ; ★ commonest cancer arising in scars ; better prognosis
Squamous cell	★ Central	Strong smoking link ; ★ cavitating lesions from a hilar bronchus; ★ PTHrP → hypercalcaemia
Large cell	Peripheral	Strong smoking link; poor chemo response ; early metastases ; poor prognosis
★ Small cell (neuroendocrine)	★ Central	Strong smoking link; ★ many paraneoplastic syndromes ; undifferentiated, very aggressive, early metastases

*Mnemonic [added]: "S & S are Sentral": Squamous and Small cell are **central**; adeno and large cell are peripheral.*

2.3 Staging (Slides 8-9)

★★ NSCLC → TNM (IASLC 8th edition, Slide 8):

	Criteria
T1	≤ 3 cm; invasion distal to lobar bronchus; superficial spreading tumour limited to bronchial wall

	Criteria
T2	> 3 to ≤ 5 cm ; main bronchus (not carina) with atelectasis/obstructive pneumonia; visceral pleura ; crosses fissure / two adjacent lobes
T3	> 5 to ≤ 7 cm ; parietal pericardium, phrenic nerve, parietal pleura, chest wall ; separate nodules in the same lobe
T4	> 7 cm ; trachea, recurrent laryngeal nerve, great vessels, diaphragm, oesophagus, vertebra, carina ; nodules in a different ipsilateral lobe
N1	Ipsilateral hilar/intrapulmonary/peribronchial
N2	Subcarinal , ipsilateral mediastinal
N3	Contralateral mediastinal/hilar; any supraclavicular or scalene
M1a	Contralateral lung nodule, pleural nodule or malignant pleural effusion
M1b	Single extrathoracic metastasis
M1c	Multiple extrathoracic metastases
Size mnemonic [added]: " 3-5-7 " cm.	

Staging:

NSCLC is staged via the primary TNM system.

T1

- Tumor ≤3 cm
- Bronchoscopically visible invasion distal to lobar bronchus
- Superficial spreading tumor of any size, invasion limited to bronchial wall

T2

- Tumor >3 cm to ≤5 cm
- Involves main bronchus, without main carinal involvement, with atelectasis and/or obstructive pneumonia of part or all of lung
- Invades visceral pleura
- Extends across fissure or involves two adjacent lobes

T3

- Tumor >5 cm to ≤7 cm
- Parietal pericardium or phrenic nerve invasion
- Invades parietal pleura
- Separate tumor nodules in the same lobe as primary tumor
- Chest wall invasion

T4

- Tumor >7 cm
- Invades trachea, recurrent laryngeal nerve, great vessels, diaphragm, esophagus, and/or vertebral body
- Involves main carina
- Separate tumor nodules in a different ipsilateral lobe

N1: Ipsilateral hilar, intrapulmonary, and/or peribronchial

N2: Subcarinal, ipsilateral mediastinal

N3: Contralateral mediastinal and hilar, ipsilateral or contralateral supraclavicular or scalene

M1a: Tumor in contralateral lung or pleural nodule or malignant pleural effusion

M1b: Single extrathoracic metastasis

M1c: Multiple extrathoracic metastases in one or more organ

Summary of Lung Cancer Staging IASLC 8th Edition

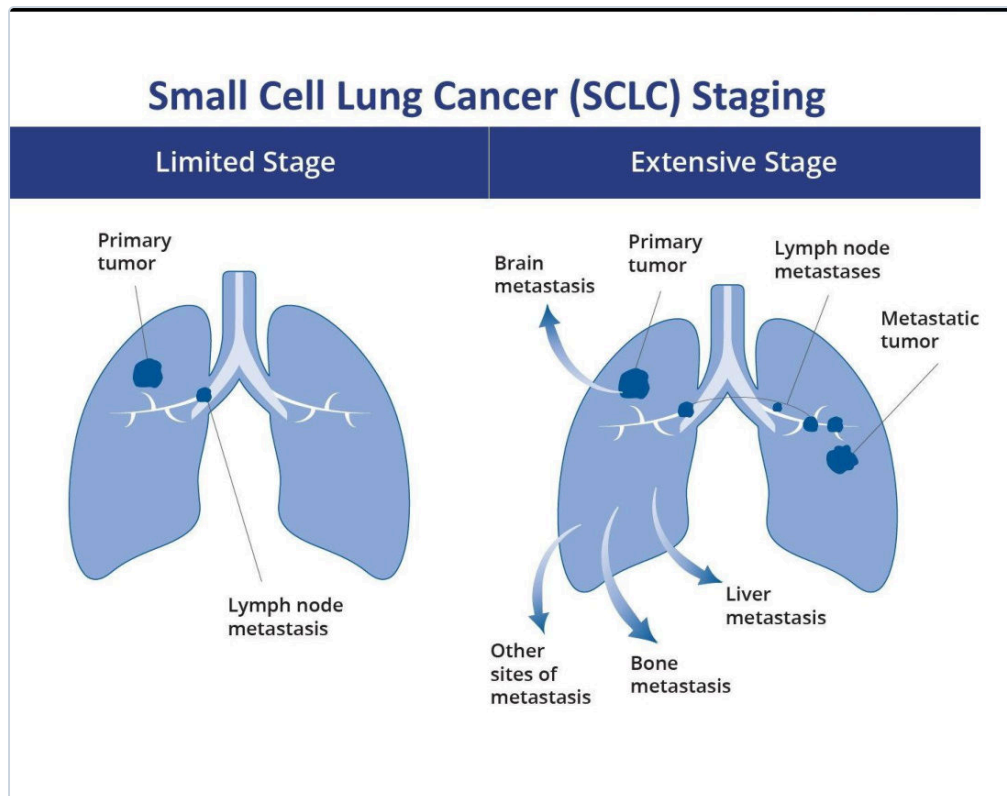
Summary of the 8th Edition of the TNM Classification for Thoracic Cancers. IASLC. 2018. www.iaslc.org/research-education/publications-resources-guidelines/summary-8th-edition-tnm-classification

CHEST

Slide 8: TNM staging summary (IASLC 8th edition)

★★ SCLC → limited vs extensive (Slide 9):

Limited	Extensive
Confined to the chest + supraclavicular nodes (not cervical or axillary)	Beyond the chest and supraclavicular nodes



Slide 9: SCLC limited vs extensive stage

🔄 **Recap: Types and staging.** NSCLC ~85% (adeno peripheral/non-smokers/scars; squamous central/cavitating/hypercalcaemia; large cell peripheral/aggressive). SCLC ~15% (central, aggressive, paraneoplastic). NSCLC → TNM (3-5-7 cm; N1 hilar, N2 mediastinal, N3 contralateral/supraclavicular; M1a pleural). SCLC → limited vs extensive.

🔗 **Link:** Tumour site and type predict the symptoms.

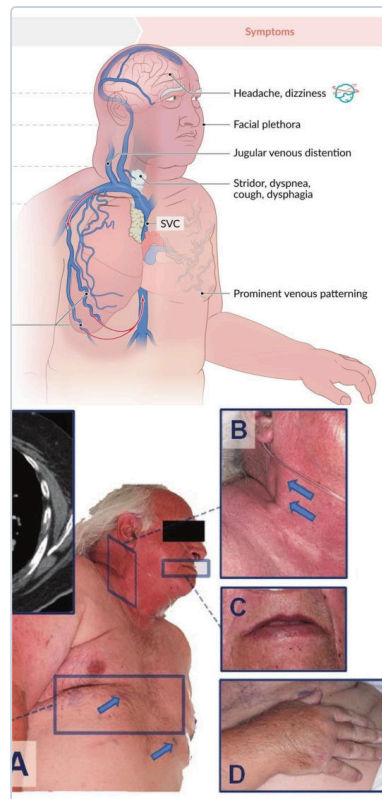
📌 **Up next: Clinical features.**

2.4 Signs & Symptoms (Slides 10-19)

★ **Five groups:** local invasion · local (pulmonary) symptoms · constitutional · metastatic · paraneoplastic.

A. ★★ Superior vena cava syndrome (Slide 11)

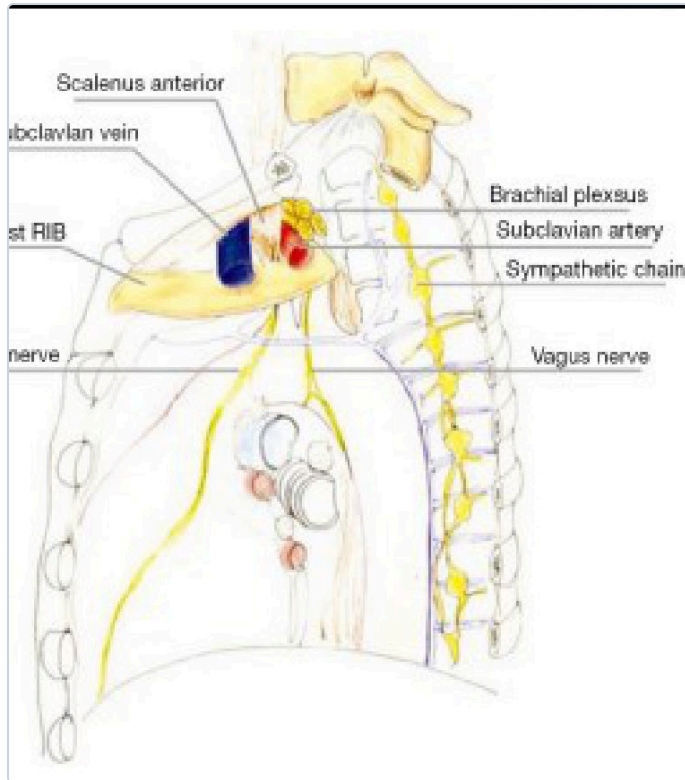
- **Obstruction of the SVC** by a **mediastinal tumour**; ★ **more common with SCLC**
- Blood diverts to the **IVC via collaterals** → **prominent veins on the upper body**; congestion above the block
- Findings: ★ **facial fullness, facial and arm oedema, dilated veins over chest, arms and face**, ★ **JVD**; hoarseness, dysphagia, post-obstructive pneumonia, dyspnoea (diaphragmatic paralysis)



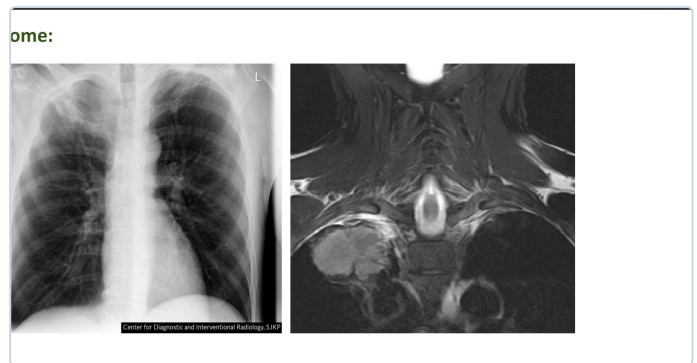
Slide 11: SVC syndrome: facial swelling and dilated chest veins

B. ★★ Pancoast tumour (Slides 12, 14)

- ★ **Apical (superior sulcus) tumour** involving **C8 and T1-T2** → ★ **shoulder pain radiating down the arm**
- **Usually NSCLC**
- **Pancoast syndrome** (mass effect):
 - **Brachial plexus** → pain, motor/sensory deficit, **wasting of small hand muscles**
 - ★ **Cervical sympathetic ganglion** → **Horner syndrome**
 - **Recurrent laryngeal nerve** → **hoarseness**
 - **Brachiocephalic vein** → **facial swelling, unilateral arm oedema**
 - **Phrenic nerve** → **hemidiaphragm paralysis**



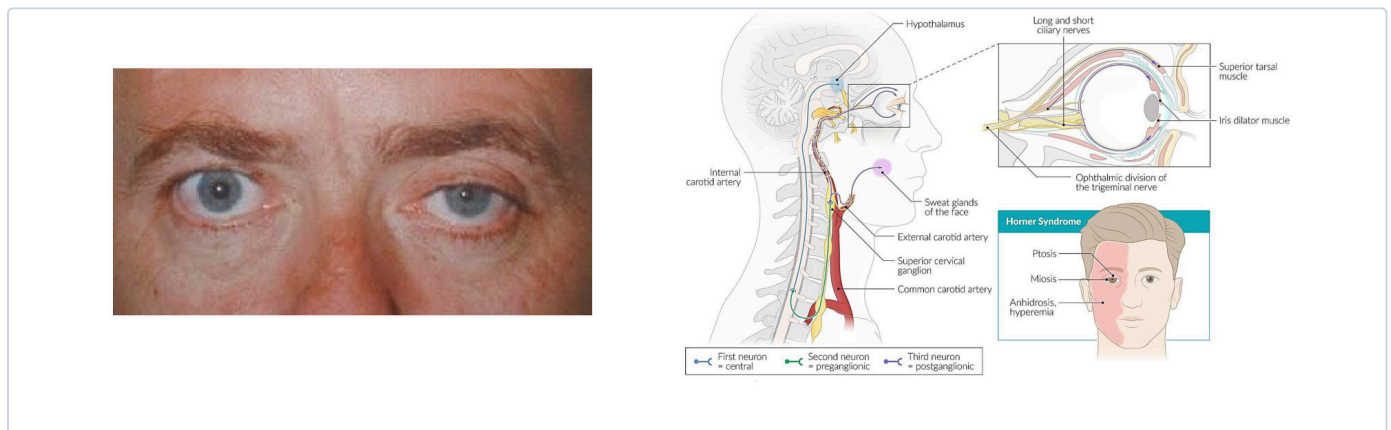
Slide 12: Pancoast tumour anatomy



Slide 14: Pancoast tumour on CXR and MRI

C. ★ Horner syndrome (Slide 13)

- Invasion of the **cervical sympathetic chain** by an apical tumour
- ★ **Ipsilateral ptosis + miosis + facial anhidrosis**



Slide 13: Horner syndrome: ptosis and miosis; sympathetic pathway

D. Pulmonary symptoms (Slide 15)

★ **Cough, haemoptysis, progressive dyspnoea, wheeze** (localised, from obstruction *[added]*), **chest pain**

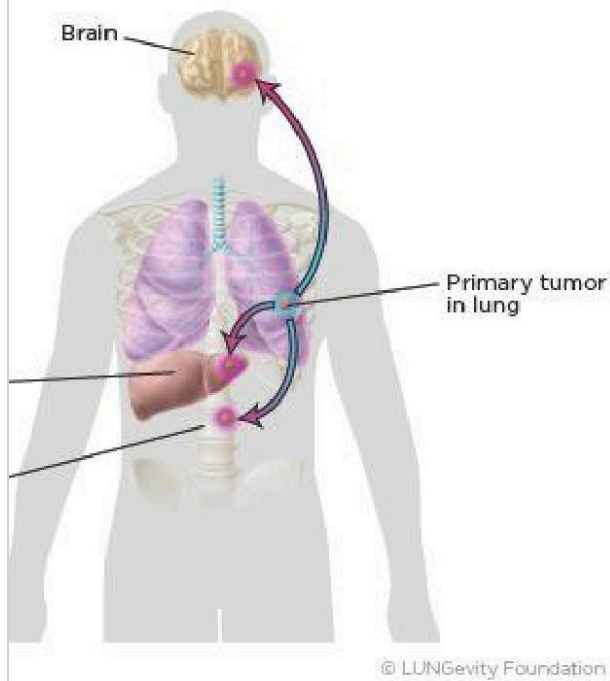
E. Extrapulmonary (Slide 16)

- ★ **Constitutional: weight loss, fever, weakness**
- **Local infiltration/compression: SVC syndrome, hoarseness** (recurrent laryngeal), **dyspnoea + raised hemidiaphragm** (phrenic), **malignant pleural effusion** (dullness, ↓ breath sounds), **post-obstructive pneumonia, dysphagia** (oesophageal compression)

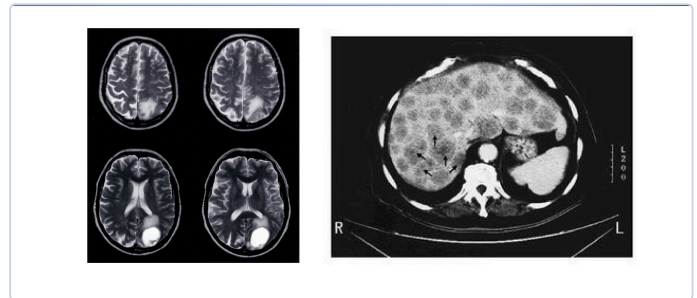
F. ★ Metastases (Slides 17-18)

Site	Features
★ Brain	Headache, seizures, focal motor deficits
Liver	Nausea, jaundice, ascites
★ Adrenal	Usually asymptomatic
★ Bone	Bone pain, ↑ ALP and calcium

Cancer Metastasis



Slide 17: Lung cancer metastatic sites

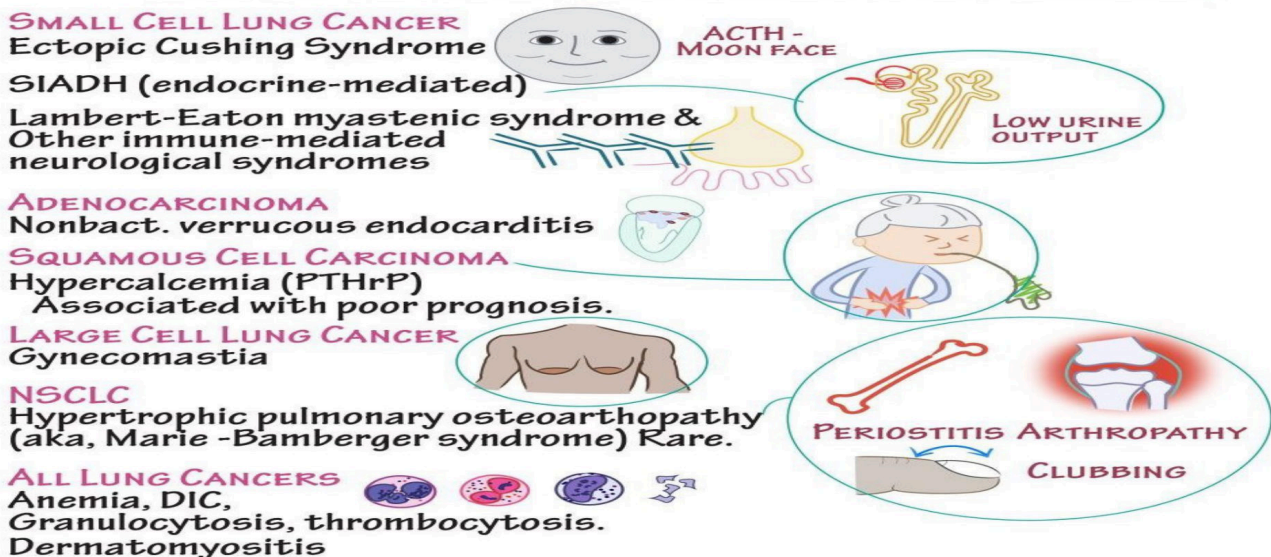


Slide 18: Brain and liver metastases on MRI/CT

G. ★★★★★ Paraneoplastic syndromes (Slide 19)

Type	Syndrome
★ Small cell	★ Ectopic Cushing (ACTH) : moon face · ★ SIADH (low urine output, hyponatraemia) · ★ Lambert-Eaton myasthenic syndrome and other immune-mediated neurological syndromes
Adenocarcinoma	Non-bacterial verrucous (thrombotic) endocarditis
★ Squamous cell	★ Hypercalcaemia (PTHrP) : poor prognosis
Large cell	Gynaecomastia
NSCLC	★ Hypertrophic pulmonary osteoarthropathy (Marie-Bamberger syndrome; rare): periostitis, arthropathy, clubbing
All lung cancers	Anaemia, DIC, granulocytosis, thrombocytosis, dermatomyositis

Paraneoplastic Syndromes & Complications



Slide 19: Paraneoplastic syndromes and complications

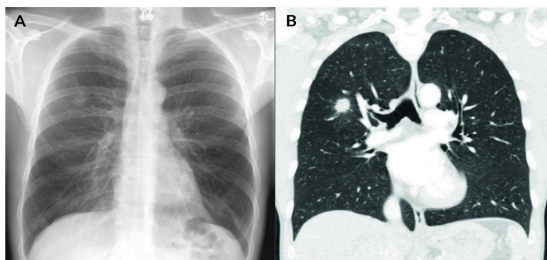
Recap: Clinical. Cough, haemoptysis, dyspnoea, weight loss. SVC syndrome (SCLC): facial swelling, dilated chest veins, JVD. Pancoast (apical NSCLC): arm pain, hand wasting, Horner, hoarseness. Metastases to brain, liver, adrenals, bone. Paraneoplastic: SCLC → ACTH, SIADH, Lambert-Eaton; squamous → PTHrP hypercalcaemia; large cell → gynaecomastia; adeno → marantic endocarditis; NSCLC → HPOA.

Link: Suspicion leads to imaging, tissue diagnosis and staging.

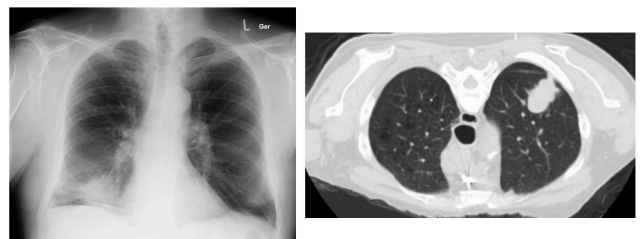
Up next: Diagnostics, treatment and screening.

2.5 Diagnostics (Slides 20-22)

1. **CXR**
2. ★ **CT with IV contrast**
3. ★ **Fibre-optic bronchoscopy with endobronchial ultrasound (EBUS)** (central lesions, mediastinal nodes)
4. **Transthoracic needle biopsy** (peripheral lesions)
5. ★ **Histopathology** (tissue diagnosis)
6. **Sputum cytology**
7. ★ **Genetic testing** (driver mutations → targeted therapy)
8. **Immunohistochemistry**
9. ★ **Staging: MRI (brain) and PET-CT**



Slide 21: CXR and CT: lung mass

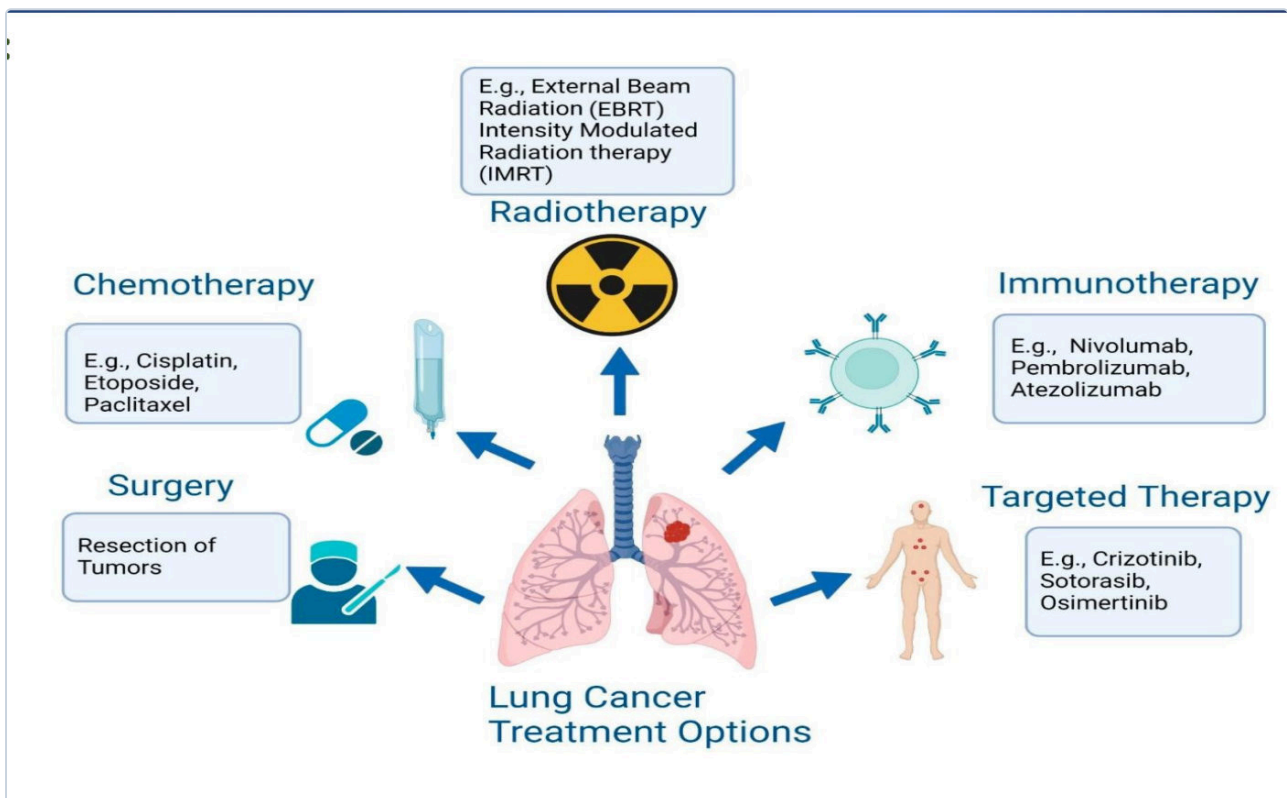


Slide 22: CXR and CT: spiculated nodule

2.6 Management (Slide 23)

Modality	Examples
Surgery	Resection (early NSCLC [added])
Radiotherapy	External beam (EBRT), IMRT
Chemotherapy	Cisplatin, etoposide, paclitaxel
★ Immunotherapy	Nivolumab, pembrolizumab, atezolizumab (PD-1/PD-L1 inhibitors [added])
★ Targeted therapy	Crizotinib (ALK), sotorasib (KRAS G12C), osimertinib (EGFR) [targets added]

[added principle: early NSCLC → surgery; locally advanced → chemoradiation ± immunotherapy; metastatic → targeted therapy if a driver mutation, else immunotherapy ± chemo. SCLC → chemotherapy ± radiotherapy; rarely surgical.]



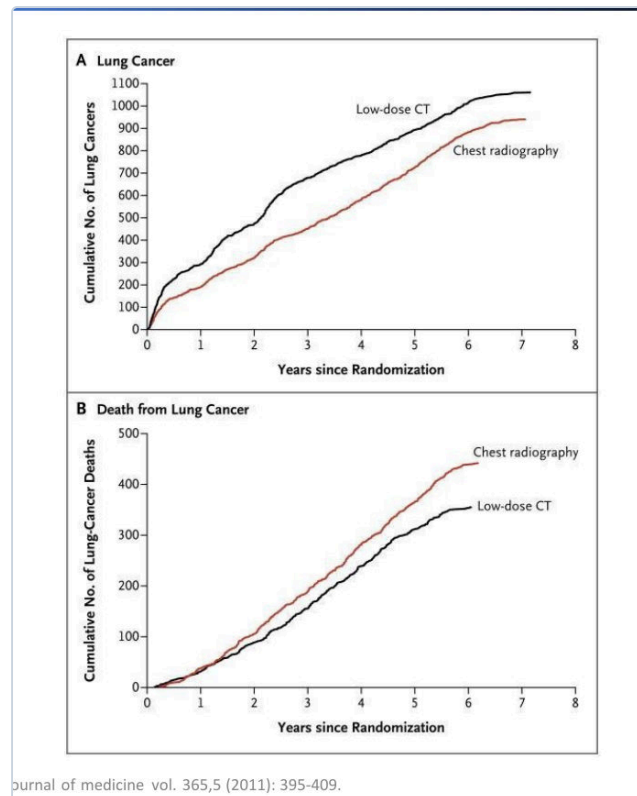
Slide 23: Lung cancer treatment options

2.7 ★★ Screening (Slide 24)

Annual low-dose CT if **all** of:

- ★ **Age 50-80**
- ★ **≥ 20 pack-year** history
- ★ **Currently smoking or quit within the past 15 years** → ★ **Reduces lung-cancer mortality by 20%** (National Lung Screening Trial, NEJM 2011)

Mnemonic [added]: "**50-80, 20 pack-years, quit < 15**".



Slide 24: NLST: fewer lung-cancer deaths with low-dose CT vs chest X-ray

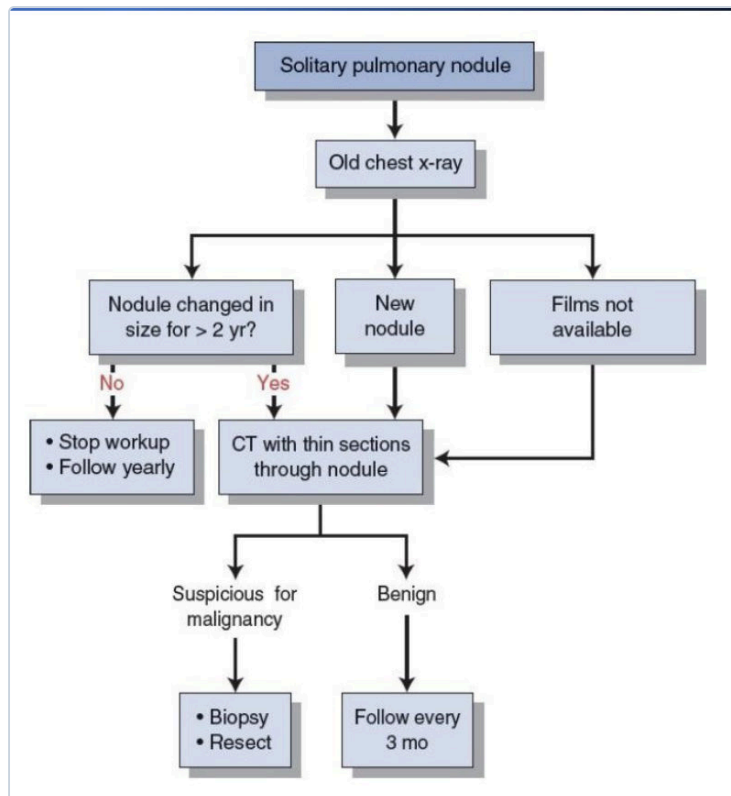
2.8 ★ Solitary Lung Nodule (Slide 25)

Approach:

1. ★ Review previous imaging
2. ★ Thin-slice CT chest
3. Assess size and features:
 - ★ Malignant features: irregular (spiculated) margins, larger size, upper-lobe location
 - ★ Benign features: calcification [added: popcorn or central/laminated patterns]

Algorithm (Slide 25): old CXR →

- ★ Unchanged for > 2 years → stop work-up, follow yearly [added: solid nodule stable 2 years = benign]
- Changed, new, or no old films → thin-section CT →
 - Suspicious → biopsy or resect
 - Benign → follow every 3 months



Slide 25: Solitary pulmonary nodule algorithm

🔁 **Recap: Diagnosis and treatment.** CT with contrast; tissue via bronchoscopy/EBUS (central) or needle biopsy (peripheral); genetics; stage with PET-CT and brain MRI. Surgery, radiotherapy, chemo, immunotherapy, targeted drugs. Screen 50-80, ≥ 20 pack-years, quit < 15 years $\rightarrow$ low-dose CT ($\sim 20\%$ mortality). Nodule: compare old films; stable > 2 years $\rightarrow$ benign; otherwise thin-slice CT $\rightarrow$ biopsy if suspicious.

🔗 **Link:** Lung cancer often presents with haemoptysis, but most haemoptysis isn't cancer.

👉 **Up next: Approach to haemoptysis.**

2.9 Approach to Haemoptysis (Slides 26-30)

★ **Key history (Slide 26):** onset · dyspnoea · cough/sputum · fever · **weight loss** · heart-disease symptoms · trauma · ★ **smoking** · **bleeding elsewhere** · ★ **medications (anticoagulants)**

Key exam (Slide 27): vital signs · **pulse oximetry** · ★ **nasal mucosa and oropharynx** (exclude upper-airway source) · heart · lungs · nodes and extremities (★ **clubbing**, cyanosis, oedema)

★★ **Differential (Slide 28):**

Group	Causes
Respiratory	★ Bronchial carcinoma, pneumonia , ★ TB (commonest worldwide) , chronic bronchitis, bronchiectasis , pulmonary oedema, Goodpasture, Wegener's (GPA)
Vascular	★ Pulmonary embolism, pulmonary hypertension (mitral stenosis) , hereditary haemorrhagic telangiectasia
Systemic	Coagulation disorders

Work-up:

General	Specific
CBC, CXR, coagulation (PT, PTT, INR), sputum analysis, ECG	CT, TB testing, echo, bronchoscopy, BMP (renal), urinalysis (pulmonary-renal syndromes <i>[added]</i>)

RESPIRATORY

- Bronchial carcinoma
- Pneumonia
- TB (worldwide) (Fig. 31)
- Chronic bronchitis
- Bronchiectasis
- Pulmonary oedema
- Goodpasture's syndrome
- Wegener's granulomatosis

VASCULAR

- Pulmonary embolism
- Pulmonary hypertension – mitral stenosis
- Hereditary haemorrhagic telangiectasia

SYSTEMIC

- Coagulation disorders

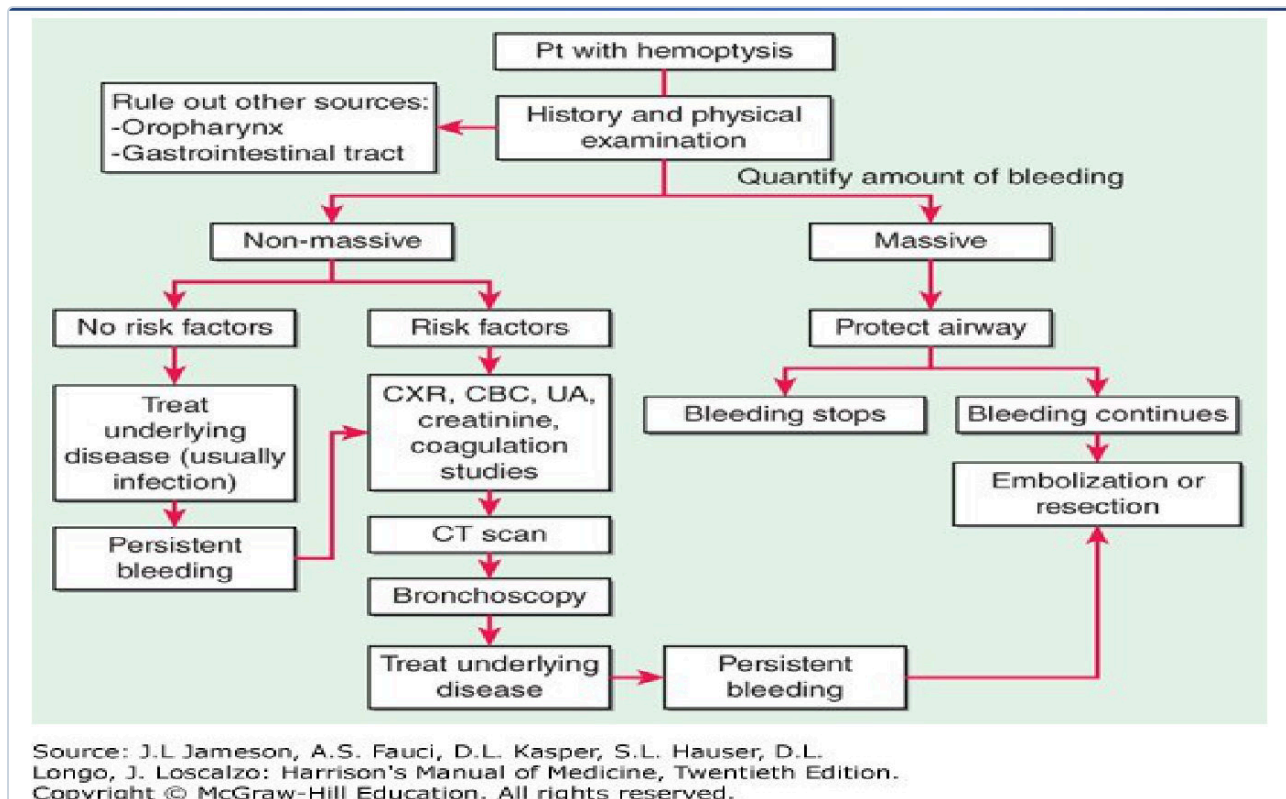
Slide 28: Haemoptysis differential and work-up

★★★★ Massive haemoptysis (Slide 29):

1. ★ **Blood group and cross-match**
2. **IV access**
3. ★ **Secure the airway** *[added: place the bleeding side down; consider selective intubation]*
4. Work-up: **CXR, coagulation, renal function, sputum**
5. ★ **Control haemorrhage: tranexamic acid, FFP, platelets if < 50, red cells** as needed
6. **Source control** (by cause)

★★ Summary algorithm (Slide 30, Harrison):

1. History and exam → ★ **rule out other sources (oropharynx, GI tract)**
2. Quantify bleeding:
 - **Non-massive, no risk factors** → treat underlying disease (usually infection) → if persistent, investigate
 - **Non-massive, risk factors** → **CXR, CBC, UA, creatinine, coagulation** → **CT** → **bronchoscopy** → treat the cause
 - ★ **Massive** → **protect the airway** → stops: work up; continues → ★ **embolisation or resection**



Slide 30: Haemoptysis algorithm: non-massive vs massive

Recap: Haemoptysis. Exclude nose, mouth and GI sources. Common causes: infection, bronchitis, bronchiectasis, TB, cancer, PE. Risk factors (smoker, age, weight loss) → CXR → CT → bronchoscopy. Massive: airway first, cross-match, IV access, TXA/FFP/platelets, then bronchial-artery embolisation or surgery.

3. High-Yield Comparison Tables

Table A: Cancer types (see §2.2 and the added diagram)

Table B: Paraneoplastic syndromes (see §2.4 G)

Table C: TNM vs SCLC staging

	NSCLC	SCLC
System	TNM	Limited vs extensive
Key cut-offs	T: 3/5/7 cm; N1 hilar, N2 mediastinal/subcarinal, N3 contralateral/supraclavicular; M1a pleural/contralateral	Limited = chest + supraclavicular nodes

Table D: Local invasion syndromes

Structure	Result
SVC	Facial/arm oedema, dilated veins, JVD
Cervical sympathetic chain	Horner: ptosis, miosis, anhidrosis
Brachial plexus (C8-T2)	Arm pain, hand-muscle wasting
Recurrent laryngeal nerve	Hoarseness

Structure	Result
Phrenic nerve	Raised hemidiaphragm, dyspnoea
Oesophagus	Dysphagia
Pleura	Malignant effusion

Table E: Nodule features

Malignant	Benign
Irregular/spiculated margins, large, upper lobe, growing	Calcified, stable > 2 years

Table F: Massive vs non-massive haemoptysis (see §2.9)

4. Rules, Exceptions, Patterns

- **Smoking = 80-90%; radon = 2nd cause.**
- **Adenocarcinoma: most common, peripheral, non-smokers and women, scars.**
- **Squamous and small cell are central.**
- **Squamous cavitates and causes hypercalcaemia (PTHrP).**
- **SCLC: SIADH, ACTH, Lambert-Eaton, SVC syndrome; staged limited vs extensive.**
- **NSCLC: TNM; Pancoast tumour is usually NSCLC.**
- **Malignant pleural effusion = M1a (stage IV).**
- **Adrenal metastases are usually silent.**
- **Screen 50-80 y, ≥ 20 pack-years, current or quit < 15 y, with annual low-dose CT.**
- **Nodule stable > 2 years → stop work-up; calcification suggests benign.**
- **Massive haemoptysis: airway first.** Persistent → embolisation or resection.
- **TB is the commonest cause of haemoptysis worldwide.**

5. Clinical Correlations

- **Smoker with hyponatraemia and confusion** → SIADH from SCLC.
- **Smoker with hypercalcaemia, normal PTH, cavitating hilar mass** → squamous cell carcinoma (Case 1).
- **Smoker with proximal weakness that improves on repeated effort** → Lambert-Eaton (anti-VGCC antibodies *[added]*).
- **Facial swelling and dilated chest veins in a smoker** → SVC syndrome → urgent imaging ± stent/radiotherapy *[added]*.
- **Shoulder pain + ptosis + small pupil + hand wasting** → Pancoast tumour.
- **Non-smoking woman with a peripheral nodule** → adenocarcinoma (EGFR testing → osimertinib *[added]*).
- **Asbestos worker with effusion** → mesothelioma or lung cancer (smoking + asbestos multiply risk *[added]*).
- **Haemoptysis + haematuria + renal failure** → Goodpasture or GPA (urinalysis + BMP).

6. Rapid Revision (Last-Day)

- Risks: **smoking 80-90%** (pack-years), passive smoking, **radon (#2)**, uranium, **asbestos**, arsenic, silica, pollution, family history, scars, radiation, fibrosis, TB, HIV.
- NSCLC ~85%** (adeno, squamous, large cell) vs **SCLC ~15%**.
- Adeno**: peripheral, most common, women/non-smokers, scars, HPOA, better prognosis.
- Squamous**: central, smokers, **cavities**, **PTHrP → hypercalcaemia**.
- Large cell**: peripheral, smokers, early metastases, poor prognosis, **gynaecomastia**.
- SCLC**: central, smokers, very aggressive, **ACTH, SIADH, Lambert-Eaton**, SVC syndrome.
- TNM: **T 3/5/7 cm**; **N1** hilar, **N2** mediastinal/subcarinal, **N3** contralateral/supraclavicular; **M1a** pleural/effusion/contralateral, **M1b** single, **M1c** multiple distant.
- SCLC: limited (chest + supraclavicular) vs extensive**.
- SVC syndrome**: facial/arm oedema, dilated veins, JVD.
- Pancoast**: apical, C8-T2 → arm pain, hand wasting, **Horner (ptosis, miosis, anhidrosis)**, hoarseness, phrenic palsy.
- Metastases: **brain** (headache, seizures), **liver** (jaundice), **adrenal** (silent), **bone** (pain, ↑ ALP, ↑ Ca).
- All cancers: anaemia, DIC, thrombocytosis, dermatomyositis.
- Dx: CXR, **CT with contrast, bronchoscopy/EBUS**, needle biopsy, histology, cytology, genetics, IHC, **PET-CT + MRI** staging.
- Rx: surgery, radiotherapy, **cisplatin/etoposide/paclitaxel, nivolumab/pembrolizumab/atezolizumab, crizotinib/sotorasib/osimertinib**.
- Screening: 50-80 y, ≥ 20 pack-years, smoking or quit < 15 y → yearly low-dose CT, -20% mortality**.
- Nodule**: compare old films; **stable > 2 y → stop, yearly**; else thin CT → suspicious → biopsy/resect; benign → every 3 months.
- Haemoptysis DDx**: cancer, pneumonia, **TB**, bronchitis, bronchiectasis, oedema, Goodpasture, GPA, **PE**, mitral stenosis, HHT, coagulopathy.
- Massive haemoptysis**: cross-match, IV access, **airway**, CXR/coagulation/renal, **TXA, FFP, platelets < 50, RBCs**, then **embolisation or resection**.

7. Exam-Style Questions

Q1 (MCQ, lecture Case 1, Slide 31). 69 ♂, 30 pack-years; 3 months of dry cough, **blood-streaked sputum**, fatigue, **5-kg weight loss**, worsening **back pain**; localised **wheeze in the right lower field**; ★ **calcium 12.3 mg/dL**, normal Hb and total protein. CXR shown (mass). Most likely diagnosis? **f. Squamous cell lung carcinoma** **Answer: f.** [reasoning added; no key on the slide] Smoker + haemoptysis + weight loss + central obstruction (localised wheeze) + **hypercalcaemia = squamous cell carcinoma with PTHrP** (back pain may be bone metastasis). **SCLC** causes SIADH/ACTH, not hypercalcaemia. **Myeloma**: normal Hb and total protein make it unlikely. TB/pneumonia don't explain the hypercalcaemia.

Q2 (MCQ, lecture Case 2, Slide 32). 50 ♂, ex-smoker (10 pack-years, quit 18 years ago), recent travel; incidental **5-mm nodule** in the right upper lung on CXR after a fall; **no previous X-rays**. Next step? **e. CT scan of the chest** **Answer: e.** Per the lecture algorithm, **no prior films → thin-section CT** to characterise the nodule. PET is for larger suspicious lesions. Bronchoscopy is premature. Interval CXR/reassurance skip characterisation. He doesn't meet screening criteria, but this is an **incidental nodule**, not screening [added].

Q3 (MCQ). A smoker presents with confusion and Na⁺ 118 mmol/L. CT shows a central hilar mass. Most likely histology? a) Adenocarcinoma b) Squamous **c) Small cell** d) Large cell **Answer: c**

(SIADH).

Q4 (MCQ). Which patient qualifies for annual low-dose CT screening? a) 45 y, 30 pack-years, current smoker **b) 62 y, 25 pack-years, quit 10 years ago** c) 70 y, 15 pack-years, current d) 65 y, 40 pack-years, quit 20 years ago **Answer: b.** (a) too young; (c) < 20 pack-years; (d) quit > 15 years.

Q5 (Short answer). List the first steps in massive haemoptysis. **Answer: Secure the airway** (bleeding side down, intubate if needed), **IV access, group and cross-match**, labs (CBC, coagulation, renal) and **CXR, control bleeding** (TXA, FFP, platelets if < 50, RBCs), then **source control: bronchial-artery embolisation or surgical resection** if it continues.

8. Exam Pitfalls & Traps

1. **Small cell → hypercalcaemia.** No: **squamous (PTHrP)**. SCLC → SIADH, ACTH, Lambert-Eaton.
2. **Adenocarcinoma is central.** No, it's **peripheral**; squamous and small cell are central.
3. **SCLC staged by TNM** (in this lecture). Use **limited vs extensive**.
4. **Most lung cancer in non-smokers is squamous.** It's **adenocarcinoma**.
5. **Malignant pleural effusion = local (T) disease.** It's **M1a** (metastatic).
6. **Screening criteria:** age **50-80**, **≥ 20 pack-years**, quit **< 15 years**. Each cut-off matters.
7. **Calcified nodule = cancer.** Calcification usually means **benign**.
8. **Stable nodule needs CT every 3 months.** If **stable > 2 years** → stop work-up, follow yearly.
9. **Massive haemoptysis: CT first. Airway and resuscitation first.**
10. **Horner triad:** ptosis, **miosis** (not mydriasis), **anhidrosis**.
11. **⚠ Slide inconsistency:** the **Pancoast** slide says "usually NSCLC", while **SVC syndrome** is "more common with SCLC". Both are correct; don't swap them.
12. **⚠ Slide typos:** "Fascial swelling" = **facial**; "myastenic" = myasthenic.